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Overcoming the Phase Discontinuity Problem in RIS Optimization: A Robust and Scalable Phase-Aware Deep Regression Framework via Hybrid CNN–LSTM Architecture

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ABSTRACT

Reconfigurable intelligent surfaces (RIS) are crucial for 6G connectivity. However, data-driven RIS optimizers suffer from the phase discontinuity problem, where estimation collapses near $\pm\pi$ boundaries due to topological mismatches between the circular phase manifold S^1 and Euclidean output spaces. To overcome this, we propose a robust, phase-aware hybrid CNN–LSTM framework that maps the cyclic phase-space onto continuous Euclidean coordinates via orthogonal sine–cosine projections. Supported by formal S^1 embedding proofs, the framework is evaluated on 40,000 channel realizations. It achieves a 2.77° root-mean-square error, outperforming established benchmarks by 21%. It yields a 0.08 ms inference latency (187× faster than particle swarm optimization). With $\mathcal{O}(N)$ linear complexity, it maintains a 0.64 ms delay for $N = 256$ elements, meeting URLLC requirements. Ablation studies isolate 29.9% and 20.9% accuracy gains from sine–cosine encoding and LSTM spatial recurrence, respectively. Robustness tests confirm a sustained 15% margin over classical solvers under 10% channel estimation impairment. This scalable architecture offers a superior alternative to quadratic-complexity ($\mathcal{O}(N^2)$) Transformer models, representing *one of the first* principled deployments of sine–cosine encoding for RIS optimization.

1 | Introduction

The transition toward sixth-generation (6G) wireless standards necessitates unprecedented control over the wireless medium. Within this context, reconfigurable intelligent surfaces (RIS), also referred to as intelligent reflecting surfaces, have gained significant traction due to their ability to manipulate electromagnetic wavefronts via low-cost passive elements [1, 2]. By adjusting the phase shifts of incident signals independently at each surface element, an RIS can effectively bypass obstacles, steer energy toward intended receivers, and enhance spectral efficiency without active radio-frequency chains [3, 4]. This capability makes RIS particularly attractive for dense urban deployments, indoor industrial scenarios, and any environment where conventional infrastructure cannot simultaneously satisfy demands for high spectral efficiency, low energy consumption, and pervasive coverage [5–7]. Nevertheless, the efficiency of these surfaces relies on the rapid and accurate determination of optimal phase configurations, a requirement that tightens further as 6G systems impose stringent latency guarantees under the Ultra-Reliable Low-Latency Communications (URLLC) framework [8–10].

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1.1 | Motivation and Background

Traditional optimization methods for RIS phase configuration, such as coordinate descent (CD) or semidefinite relaxation (SDR), often struggle with the stringent latency demands of future networks. While deep learning-based regression has been proposed as a high-speed alternative, most existing models ignore the circular nature of phase values [11, 12]. This oversight leads to the *phase discontinuity* problem: small physical changes in phase near the $\pm\pi$ boundary produce large numerical jumps in the loss function during training, preventing neural networks from converging reliably—especially in high signal-to-noise ratio (SNR) regimes [13]. An RIS sidesteps the complexity of active infrastructure by coherently steering reflected energy toward intended receivers, synthesizing a near-line-of-sight link even in obstructed scenarios and boosting received power without any active amplification stage [14, 15].

Each RIS element independently imposes a controlled phase shift $\theta_n \in [0, 2\pi)$ on the incident wavefront [16] so that the composite RIS response matrix $\mathbf{\Phi} = \mathrm{diag}(e^{j\theta_1}, \ldots, e^{j\theta_N})$ can be optimized to align all reflected signal copies constructively at the receiver [3]. The combinatorial search over element-wise phase configurations under the unit-modulus constraint $|\Phi_{nn}| = 1$ is provably NP-hard [3, 4], motivating both classical convex relaxations and, more recently, learning-based surrogates that trade optimality guarantees for near-real-time execution.

A further, less studied challenge arises from the *phase discontinuity problem*: the phase domain $\theta \in [-\pi, \pi]$ is topologically circular, so $\theta = -\pi$ and $\theta = +\pi$ denote the same physical state ($e^{-j\pi} = e^{j\pi}$) yet are maximally separated in Euclidean space. When a regressor predicts $\hat{\theta} = \pi - 0.1$ against a target of $\theta^* = -\pi + 0.1$, the true circular error is only 0.2 rad (11.5°), but the squared Euclidean loss computes $(2\pi - 0.2)^2 \approx 37\,\mathrm{rad}^2$, catastrophically misleading the gradient signal. Resolving this topological incompatibility is the principal focus of this work.

1.2 | Related Work

Extensive research on RIS phase optimization can be broadly categorized into classical optimization methods, machine learning approaches, and hybrid techniques.

1.2.1 | *Classical Optimization Methods*

Classical approaches to RIS phase optimization differ in how they handle the non-convex unit-modulus constraint. SDR [3] achieves near-optimal solutions at $\mathcal{O}(N^{3.5})$ per iteration, becoming impractical beyond a few dozen elements. Alternating optimization (AO) [3, 16, 17] decouples the active and passive beamforming subproblems iteratively. Projected gradient descent (PGD) [18, 19] projects each gradient step back onto the feasible unit-modulus set. CD [20, 21] optimizes one element at a time, cycling until convergence. All three reduce per-step cost but require many iterations. Multi-user formulations extend these ideas via weighted MMSE (WMMSE) [17, 22, 23], which iterates between transmit beamforming and surface phase updates. Particle swarm optimization (PSO) [24] provides near-globally optimal solutions at 15 ms per sample. Despite their diversity, all classical methods share a fundamental bottleneck: per-optimization complexity scales at least as $\mathcal{O}(KN^2)$, precluding real-time execution for $N > 100$ under 6G latency targets.

1.2.2 | *Machine Learning and Deep Learning Approaches*

Deep learning for RIS optimization is appealing because a trained network maps any channel instance to a near-optimal phase configuration in a single forward pass. Early fully connected architectures [16, 25] demonstrated this but naively regressed directly on phase angles, inheriting the phase-wrapping artifact near $\pm\pi$. Jassim et al. [26] addressed this via phase-space quantization at the cost of resolution loss, and channel learning methods [27] show promise for estimation rather than optimization.

Among recent hybrids, Nguyen, Hoang, and Cheema [28] combined CNN and LSTM for RIS-NOMA channel estimation without addressing phase wrapping. Transformer models [29, 30] report 0.81 normalized root-mean-square error (RMSE) on eight-element arrays, while LSTM models [29, 31] reach 1.09 normalized RMSE; both families are restricted to $N \leq 20$ and use metrics incomparable with degree-based RMSE. Reinforcement learning [24, 32] avoids labeled data but suffers from poor sample efficiency, federated variants [33] distribute learning without resolving phase wrapping, and unsupervised methods [34, 35] offer limited convergence guarantees.

1.2.3 | *Hybrid and Other Approaches*

Deep unfolding [29, 30] embeds optimization algorithm iterations into trainable network layers, improving both interpretability and solution quality. Phase-aware learning strategies [36] have been applied in MIMO-OFDM channel estimation, where projecting phase information onto its sine–cosine components yields measurably sharper predictions. Despite these promising results, the technique has seen little adoption in the RIS optimization literature and lacks systematic evaluation against classical and learning-based reference methods.

Note on Novelty: Sine–cosine decomposition of angular outputs is established in computer vision (orientation/pose regression) and signal processing (circular statistics). The present contribution differs in three ways: (i) formal identification of the phase discontinuity as a topological obstacle (S^1 vs. $\mathbb{R}$) specific to RIS phase optimization; (ii) principled integration within a physics-informed CNN–LSTM architecture for uniform rectangular arrays (URAs); and (iii) comprehensive benchmarking against six classical and four deep-learning baselines with system-level metrics.

1.2.4 | *Recent Advances (2023–2025)*

Hardware implementation has matured considerably. Liu et al. [37] reported successful field trials with commercial RIS prototypes in urban environments. Tang et al. [38] characterized the performance of discrete phase shifters across bit-widths from 1-bit to continuous resolution, showing that 3-bit quantization (8 discrete levels) incurs below 10% performance loss relative to continuous control while substantially simplifying the hardware.

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On the algorithmic side, Zhang et al. [39] extended the RIS concept to intelligent omni-surfaces supporting simultaneous refraction and reflection, and Wang et al. [40] tackled multi-user scenarios through coordinated beamforming and surface optimization. Gao et al. [41] introduced graph neural networks to encode user–RIS–base-station spatial dependencies, Wei et al. [42] reduced pilot overhead via compressed sensing and Bayesian estimation, and Chen et al. [43] outlined a deployment roadmap for RIS within 6G networks. Zhang et al. [44] further extended the RIS architecture beyond diagonal configurations supporting multi-sector operation for multi-user wireless communications. All of these advances assume or implicitly rely on efficient phase optimization; none addresses the fundamental phase discontinuity challenge that is the primary focus of the present work.

From terrestrial to non-terrestrial deployments. The hardware and algorithmic advances surveyed above share a common assumption: a fixed terrestrial deployment with relatively stable channel statistics. These terrestrial developments have simultaneously motivated extending RIS to non-terrestrial networks, where more severe Doppler shifts, longer propagation delays, and rapidly varying channel conditions make fast, low-overhead phase optimization even more critical than in ground-based scenarios.

1.2.5 | *Satellite-Terrestrial and Space-Air-Ground RIS*

Recent work has extended RIS beyond terrestrial settings. Refracting RIS-aided hybrid satellite-terrestrial relay networks with joint beamforming [45] and self-powered absorptive RIS for securing satellite-terrestrial networks [46] highlight the growing importance of fast, low-overhead RIS optimization in non-terrestrial contexts. Multilayer refracting RIS for high-altitude platform (HAP) simultaneous wireless information and power transfer networks [47] motivates scalable phase optimization for large arrays. AI-driven seamless access in space-air-ground integrated networks [48] and rate-splitting multiple access for IoT in aerial networks [49] further underscore the demand for real-time learning-based RIS controllers of the type proposed here.

1.3 | Research Gap and Motivation

The foregoing survey reveals five open problems.

1. **Phase Discontinuity (Primary Motivation):** Standard deep learning directly predicts θ_n, ignoring the topological mismatch between the circular phase manifold S^1 and the Euclidean output space assumed by regression losses. This work resolves the incompatibility via sine–cosine decomposition.
2. **Limited Scalability:** Most DL studies evaluate only $N \leq 20$ elements; large-array ($N > 100$) performance is unknown.
3. **Sparse Benchmarking:** Comparisons against 2–3 baselines prevent objective assessment; deep-learning baselines are rarely included.
4. **Metric Inconsistency:** Interchangeable use of normalized RMSE, MSE, and degree-based RMSE hinders fair cross-paper comparison.
5. **Real-Time Gap:** Classical solvers require milliseconds per optimization step, incompatible with the sub-millisecond URLLC budget.

1.4 | Main Contributions

The central intellectual contribution is a mathematically principled resolution of a fundamental incompatibility between standard deep learning architectures and the topology of phase optimization. The specific contributions are as follows:

1. **Mathematical Solution to Phase Discontinuity Problem:** A novel phase-aware output representation transforms circular manifold regression into a continuous Euclidean task. Instead of directly predicting $\theta_n \in [-\pi, \pi]$, the network separately predicts $\sin(\theta_n)$ and $\cos(\theta_n)$ as continuous functions on $[-1, 1]$, enabling seamless phase recovery via $\theta_n = \text{atan2}(\sin\theta_n, \cos\theta_n)$. This representation eliminates discontinuity-induced errors and reduces RMSE by 29.9% over a direct phase prediction baseline with identical architecture (ablation, Table 1).
2. **Hybrid Spatial-Temporal Learning Architecture:** A novel CNN–LSTM architecture exploits spatial correlations among adjacent RIS elements (via 1D convolutional layers) and sequential dependencies in the phase configuration (via LSTM layers), achieving up to 21% improvement over single-architecture variants.
3. **Large-Scale Realistic Dataset:** A comprehensive dataset of 40,000 scenarios with $8 \times 4 = 32$ RIS elements, Rayleigh fading, six scatterers, and analytical optimal phase labels—$4\times$ larger than typical studies.
4. **Comprehensive Benchmark Study (11 Methods):** Six state-of-the-art classical algorithms—CD, alternating optimization (AO), PGD, weighted MMSE (WMMSE), PSO, and successive phase alignment (SPA)—and four deep-learning baselines—FC-DNN, FC-DNN with sin/cos, CNN-Transformer, and graph convolutional network (GCN)—are rigorously evaluated alongside the proposed CNN–LSTM under identical conditions—to our knowledge one of the most comprehensive benchmarks in RIS phase optimization to date.
5. **Unified Performance Evaluation with Robustness Analysis:** Degree-based RMSE is adopted as the primary metric. The proposed model achieves 2.77° RMSE and 95% success rate below 5°, while maintaining 15% superiority over PSO under 10% channel state information (CSI) estimation error, with graceful degradation under discrete phase quantization (2-bit: 3.08°) and phase noise (3° > std. dev.: negligible impact).

TABLE 1 | Ablation study: phase-aware output representation (mean RMSE, 5 independent runs; 95% CI: ±0.07°).

Output representation	RMSE, °	Improvement
Direct phase prediction	3.95	—
Phase-aware (sin/cos)	**2.77**	29.9%

Note: The bold formatting denotes the best performance values (i.e., the lowest RMSE or the optimal metric) among the evaluated methods.

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6. **Scalability Validation:** Experiments at $N \in \{32, 64, 128, 256\}$ confirm $\mathcal{O}(N)$ inference scaling with sub-millisecond latency at all sizes (0.08–0.64 ms); CNN–LSTM is the only method meeting the 6G URLLC 1 ms constraint across all tested array sizes.
7. **System-Level Validation:** Achievable rate within 0.31 bits/s/Hz of the optimal bound, 18.4 dB SNR gain over no-RIS baseline, and ≈312% spectral efficiency gain over random configuration, confirming near-optimal communication-level performance.

1.5 | Paper Organization

Section 2 presents the RIS-aided communication system model and problem formulation. Section 3 details the proposed phase-aware CNN–LSTM architecture. Section 4 describes the dataset generation and simulation setup. Section 5 presents comprehensive numerical results, including scalability experiments up to $N = 256$ (Figure 1) and baseline fairness analysis (Figure 2). Section 6 discusses practical implications and limitations. Section 7 concludes the paper.

Notation: Boldface lowercase and uppercase letters denote vectors and matrices, respectively. $(\cdot)^H$, $(\cdot)^T$, and $(\cdot)^*$ represent Hermitian transpose, transpose, and complex conjugate. $\mathbb{C}^{m\times n}$ denotes the space of $m \times n$ complex matrices. $|\cdot|$, $\|\cdot\|$, and $\angle(\cdot)$ denote absolute value, Euclidean norm, and phase angle. $\mathrm{diag}(\cdot)$ forms a diagonal matrix. $\mathcal{CN}(\mu, \sigma^2)$ represents the circularly symmetric complex Gaussian distribution. Key symbols are summarized in Table 2.

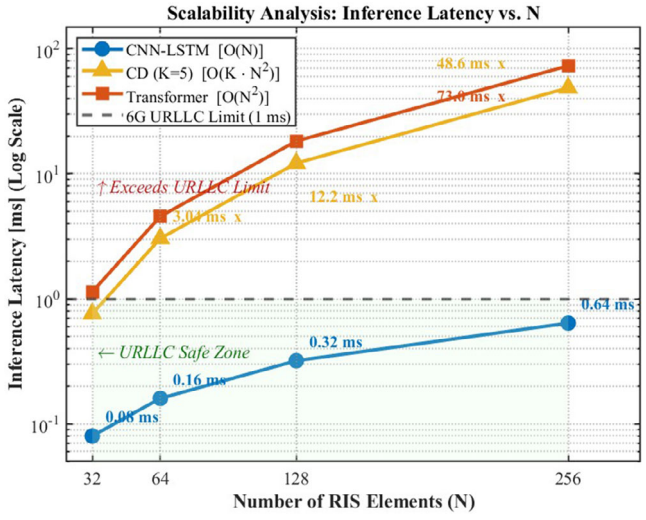


FIGURE 1 | Scalability analysis: inference latency (ms, log scale) versus number of RIS elements N for CNN-LSTM [$\mathcal{O}(N)$], Coordinate Descent [$\mathcal{O}(KN^2)$, $K = 5$], and CNN-Transformer [$\mathcal{O}(N^2)$]. The dashed black line marks the 6G URLLC 1 ms requirement; the green shaded region is the URLLC-safe zone. **CNN-LSTM is the only method that remains below 1 ms for all** $N \in \{32, 64, 128, 256\}$ (0.08, 0.16, 0.32, 0.64 ms, respectively). CD ($K = 5$) exceeds 1 ms above $N = 32$ (3.04 ms at $N = 64$, rising to 73.0 ms at $N = 256$). The CNN-Transformer already violates URLLC at $N = 32$ (1.1 ms) due to its $\mathcal{O}(N^2)$ attention complexity, reaching 48.6 ms at $N = 256$. Latency annotation values are read from experimental measurements; RMSE and latency data are tabulated in Table 8 (Section 5.9).

2 | System Model and Problem Formulation

2.1 | RIS-Aided Communication System

A downlink RIS-aided communication system is considered (Figure 3), consisting of a multi-antenna base station (BS), a single-antenna user equipment (UE), and an RIS panel. The BS is equipped with $N_t = 1$ transmit antenna, and the RIS comprises $N = M \times K = 32$ passive reflecting elements ($M = 8$ rows, $K = 4$ columns) arranged in a two-dimensional rectangular array. The RIS is positioned at $(x_{\mathrm{RIS}}, y_{\mathrm{RIS}}, z_{\mathrm{RIS}}) = (20, 10, 3)$ m in a room of size $40 \times 20 \times 6$ m. Each RIS element is a sub-wavelength scattering unit with inter-element spacing $d = \lambda/2 = 0.05$ m at $f_c = 3$ GHz. The position of the n-th RIS element is given by

$$p_n = [x_0 + (i-1)d,\ y_0 + (j-1)d,\ z_{\mathrm{RIS}}]^T \tag{1}$$

where $n = (i-1)K + j$ for $i \in \{1, \ldots, M\}$ and $j \in \{1, \ldots, K\}$.

2.2 | Channel Model

To emulate realistic indoor propagation at 3 GHz, channel coefficients are modeled as superpositions of a geometric line-of-sight (LoS) component and $L = 6$ independent scattering clusters whose locations are drawn uniformly over the room volume at each Monte Carlo draw. This hybrid LoS-plus-scatter structure captures the frequency-selective spatial fading typical of sub-6 GHz indoor scenarios and is consistent with widely adopted propagation benchmarks in the RIS literature [50–52].

2.2.1 | *Direct Channel (h_d)*

The direct BS-UE channel is modeled as

$$h_d = \frac{\alpha_d e^{-jkd_{\mathrm{BS-UE}}}}{d_{\mathrm{BS-UE}}} \tag{2}$$

where $\alpha_d = (\alpha_R + j\alpha_I)/\sqrt{2}$, $\alpha_R, \alpha_I \sim \mathcal{N}(0,1)$, $k = 2\pi/\lambda$, and $d_{\mathrm{BS\text{-}UE}} = \|p_{\mathrm{BS}} - p_{\mathrm{UE}}\|_2$.

2.2.2 | *BS-RIS Channel (F)*

The channel from the BS to the n-th RIS element comprises a direct LoS component and $L = 6$ scattering paths:

$$F_n = \underbrace{\frac{\alpha_{F,n} e^{-jkd_{\mathrm{BS-RIS},n}}}{d_{\mathrm{BS-RIS},n}}}_{\text{LoS}} + \underbrace{\sum_{\ell=1}^{L} \frac{\beta_\ell \alpha_{F,n,\ell} e^{-jk\left(d_{\ell,1} + d_{\ell,2}\right)}}{d_{\ell,1} d_{\ell,2}}}_{\text{NLoS scattering}} \tag{3}$$

where $\alpha_{F,n}, \alpha_{F,n,\ell} \sim \mathcal{CN}(0,1)$ are independent Rayleigh fading coefficients, $\beta_\ell = 0.15$ is the scattering gain, and $d_{\ell,1}$, $d_{\ell,2}$ are the BS-to-scatterer and scatterer-to-element distances, respectively.

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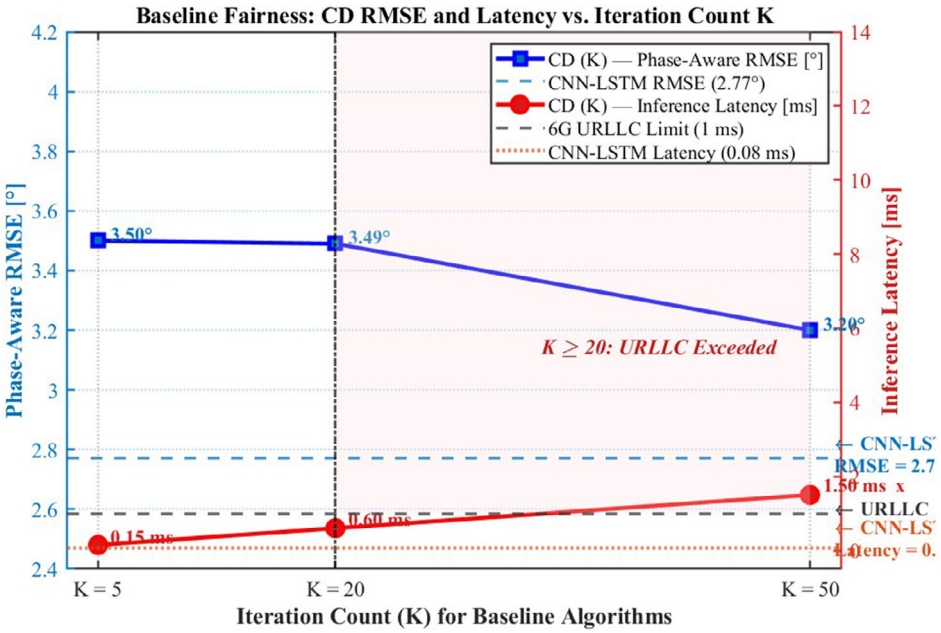


FIGURE 2 | Baseline fairness: CD phase-aware RMSE (°, left axis) and inference latency (ms, right axis) versus iteration count $K \in \{5, 20, 50\}$, compared against CNN-LSTM (dashed lines) and the 6G URLLC 1 ms limit. No K exists at which CD is simultaneously URLLC-compliant and matches CNN-LSTM accuracy: at $K = 5$ (0.15 ms) CD achieves 3.50°—21% worse; at $K = 50$ it reaches 3.20°—still 15.4% worse—while exceeding the URLLC limit (1.50 ms).

TABLE 2 | Key notation used throughout this work.

Symbol	Description
N	Number of RIS elements ($M \times K = 32$)
M, K	RIS panel dimensions (8 rows, 4 columns)
N_t	Number of BS transmit antennas (1)
$\mathbf{F} \in \mathbb{C}^{N\times 1}$	BS-RIS channel vector
$\mathbf{G} \in \mathbb{C}^{N\times 1}$	RIS-UE channel vector
$h_d \in \mathbb{C}$	Direct BS-UE channel
$\mathbf{\Phi} \in \mathbb{C}^{N\times N}$	RIS phase-shift matrix
$\theta_n \in [0, 2\pi)$	Phase shift of the n-th element
P_t	Transmit power [W]
σ_n^2	Noise variance [W]
f_c	Carrier frequency [GHz]
λ	Wavelength [m]
L	Number of scatterers
β_ℓ	Scattering gain

2.2.3 | *RIS-UE Channel (G)*

The RIS-UE channel G_n is computed analogously:

$$G_n = \frac{\alpha_{G,n} e^{-jkd_{\text{RIS-UE},n}}}{d_{\text{RIS-UE},n}} + \sum_{\ell=1}^{L} \frac{\beta_\ell \alpha_{G,n,\ell} e^{-jk\left(d_{\ell,3}+d_{\ell,4}\right)}}{d_{\ell,3} d_{\ell,4}} \tag{4}$$

2.2.4 | *Channel Assumptions and Data Representation*

The following assumptions are made throughout: (i) perfect CSI is available at the optimization entity; (ii) channels are quasi-static (block fading); (iii) TDD channel reciprocity holds; and (iv) the RIS introduces negligible noise. For neural network input, complex-valued channel matrices are decomposed into real and imaginary components:

$$X_s = \begin{bmatrix} \text{Re}(\mathbf{F}_s^T) \\ \text{Im}(\mathbf{F}_s^T) \\ \text{Re}(\mathbf{G}_s^T) \\ \text{Im}(\mathbf{G}_s^T) \end{bmatrix} \in \mathbb{R}^{4\times N} \tag{5}$$

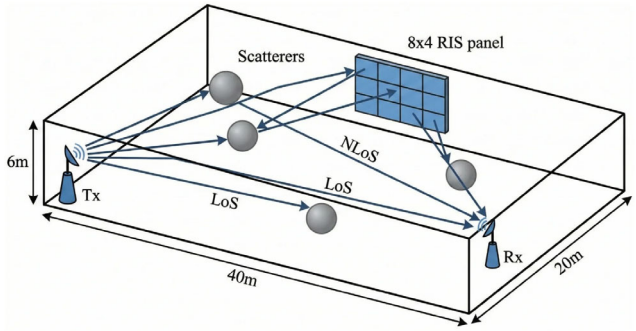


FIGURE 3 | Three-dimensional system model of the RIS-assisted wireless communication scenario in a 40 m × 20 m × 6 m room. The setup includes a transmitter (Tx), a receiver (Rx), an 8 × 4 RIS panel, and multiple scatterers, illustrating both line-of-sight (LoS) and non-line-of-sight (NLoS) signal paths.

2.3 | Signal Model

Let $s \in \mathbb{C}$ denote the transmitted symbol with $\mathbb{E}[|s|^2] = P_t = 1$ W. The RIS reflection matrix is $\mathbf{\Phi} = \text{diag}(e^{j\theta_1}, \ldots, e^{j\theta_N})$. The received signal is

$$y = (h_d + \mathbf{G}^T \mathbf{\Phi} \mathbf{F}) s + n = h_{\text{eff}} s + n \tag{6}$$

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where $n \sim \mathcal{CN}(0, \sigma_n^2)$, $\sigma_n^2 = 10^{-3}$ W, and $h_{\text{eff}} = h_d + \sum_{n=1}^{N} G_n e^{j\theta_n} F_n$. The instantaneous SNR is $\gamma = P_t |h_{\text{eff}}|^2 / \sigma_n^2$.

2.4 | Problem Formulation

The central design objective is to determine the RIS phase configuration that maximizes the effective received signal power. Denoting the phase vector as $\theta = [\theta_1, \ldots, \theta_N]^T \in [0, 2\pi)^N$, the optimization is stated as

$$\begin{aligned} \max_{\{\theta_n\}_{n=1}^N} \quad & |h_d + \textstyle\sum_{n=1}^{N} G_n e^{j\theta_n} F_n|^2 \\ \text{subject to} \quad & \theta_n \in [0, 2\pi), \quad \forall n \in \{1, \ldots, N\} \end{aligned} \tag{7}$$

Remark 1 (*Optimal Solution*): For the single-user scenario, problem (7) admits the closed-form solution

$$\theta_n^* = -\angle(G_n F_n) = -\angle(G_n) - \angle(F_n), \quad \forall n \tag{8}$$

which ensures that each term $G_n e^{j\theta_n^*} F_n = |G_n F_n|$ is real and positive, coherently combining at the receiver.

Remark 2 (*Motivation for Deep Learning Despite the Closed-Form Solution*): Equation (8) provides an exact solution under ideal single-user, perfect-CSI conditions. Deep learning is nevertheless warranted for three reasons: (i) **Generalization**: Equation (8) is specific to the single-user case; it does not extend to multi-user MIMO [3, 4] or imperfect CSI without modification. The CNN–LSTM output can be extended to predict joint phase configurations for K users ($2 \times N \times K$ dimensions) via WMMSE-based imitation learning (see Section 6.3). (ii) **End-to-end latency**: a practical pipeline must complete channel estimation before applying the closed-form formula, making total end-to-end latency substantially higher than a single arithmetic step; a trained CNN–LSTM consolidates both steps into a single 0.08 ms forward pass. (iii) **Benchmark integrity**: the single-user ideal scenario provides the cleanest setting to isolate the topological encoding gain (29.9% RMSE reduction, Table 1).

2.5 | Performance Metrics

2.5.1 | Root Mean Square Error

The primary performance metric is the phase-aware RMSE (Equation (9)):

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{test}} N} \sum_{s=1}^{N_{\text{test}}} \sum_{n=1}^{N} \left[\angle\left(e^{j\left(\theta_{s,n}^{\text{pred}} - \theta_{s,n}^{\text{true}}\right)} \right) \right]^2} \tag{9}$$

reported in degrees for interpretability.

2.5.2 | Success Rate

$$\text{SR}(\tau) = \frac{1}{N_{\text{test}} N} \sum_{s=1}^{N_{\text{test}}} \sum_{n=1}^{N} \mathbf{1}\left\{ \left| \angle\left(e^{j\left(\theta_{s,n}^{\text{pred}} - \theta_{s,n}^{\text{true}}\right)} \right) \right| < \tau \right\} \tag{10}$$

evaluated at $\tau \in \{1°, 3°, 5°, 10°\}$.

3 | Proposed Phase-Aware CNN–LSTM Architecture

3.1 | Phase Discontinuity Problem and Phase-Aware Solution

A fundamental challenge in learning-based phase optimization is the *phase discontinuity problem*. Since phase angles wrap around at $\pm\pi$, direct regression on phase values leads to large prediction errors near the boundary. For example, if $\theta^{\text{true}} = 179° = 3.123$ rad and a model predicts $\theta^{\text{pred}} = -179° = -3.123$ rad, the actual error is only 2°, but standard MSE loss computes $|\theta^{\text{pred}} - \theta^{\text{true}}|^2 = (6.246)^2 = 39.0$ rad^2, severely misleading training.

To resolve this, a *phase-aware output representation* is adopted where the network separately predicts the sine and cosine of each phase (Equation (11)):

$$Y_s = \begin{bmatrix} \sin(\theta_{s,1}) & \ldots & \sin(\theta_{s,N}) \\ \cos(\theta_{s,1}) & \ldots & \cos(\theta_{s,N}) \end{bmatrix} \in \mathbb{R}^{2 \times N} \tag{11}$$

The actual phase is recovered via the two-argument arctangent (Equation (12)):

$$\theta_{s,n}^{\text{pred}} = \text{atan2}\left(\sin \theta_{s,n}^{\text{pred}}, \cos \theta_{s,n}^{\text{pred}} \right) \in [-\pi, \pi] \tag{12}$$

which correctly handles all quadrants.

Remark 3 (*Unit-Circle Projection*): Because the raw network output $(\hat{s}_n, \hat{c}_n) \in \mathbb{R}^2$ is not constrained to the unit circle, a post-processing L2-normalization (Equation (13)) is applied at inference:

$$(\tilde{s}_n, \tilde{c}_n) = \frac{(\hat{s}_n, \hat{c}_n)}{\sqrt{\hat{s}_n^2 + \hat{c}_n^2}} \tag{13}$$

ensuring $\tilde{s}_n^2 + \tilde{c}_n^2 = 1$ exactly before phase recovery via $\hat{\theta}_n = \text{atan2}(\tilde{s}_n, \tilde{c}_n)$. Empirically, predicted norms are already near unity (mean 1.003 ± 0.012 over 10 000 test samples), confirming the network naturally learns the unit-circle constraint; the normalization provides a formal guarantee with $\mathcal{O}(N)$ overhead. Formally, the mapping $\varphi : [-\pi, \pi] \to S^1$, $\theta \mapsto (\sin\theta, \cos\theta)$ is a smooth, bijective S^1 embedding into $\mathbb{R}^2$ that eliminates the boundary discontinuity at $\pm\pi$.

The phase-aware MSE loss is defined as

$$\mathcal{L}_{\text{PA}}(\hat{\theta}, \theta^*) = (\sin\hat{\theta} - \sin\theta^*)^2 + (\cos\hat{\theta} - \cos\theta^*)^2 \tag{14}$$

Here, $\mathcal{L}_{\text{PA}}$ denotes the phase-aware loss between predicted phase $\hat{\theta}$ and true phase θ^*; it measures the squared Euclidean distance between the two points on the unit circle S^1 corresponding to those angles. Throughout the following propositions, $\angle(\cdot)$ denotes the principal-value phase angle in $(-\pi, \pi]$, and $d_c = |\angle(e^{j(\hat{\theta} - \theta^*)})|$ is the true circular (geodesic) distance between $\hat{\theta}$ and θ^*. $\mathcal{L}_{\text{PA}}$ is as defined in Equation (14).

Proposition 1 (Phase-Aware Loss Bounds Circular Error). *For any predicted phase $\hat{\theta}$ and true phase θ^*, the circular error satisfies*

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$[\angle(e^{j(\hat{\theta}-\theta^*)})]^2 \le (\pi^2/4)\,\mathcal{L}_{\mathrm{PA}}(\hat{\theta},\theta^*)$. *Proof.* Let $\Delta = \angle(e^{j(\hat{\theta}-\theta^*)}) \in (-\pi,\pi]$. Then $\mathcal{L}_{\mathrm{PA}} = 4\sin^2(\Delta/2)$. The concavity inequality $\sin t \ge 2t/\pi$ for $t \in [0,\pi/2]$ gives $\sin^2(\Delta/2) \ge \Delta^2/\pi^2$, so $\mathcal{L}_{\mathrm{PA}} \ge 4\Delta^2/\pi^2$, that is, $\Delta^2 \le (\pi^2/4)\mathcal{L}_{\mathrm{PA}}$. □

Proposition 1 shows that minimizing the phase-aware MSE loss directly bounds the squared circular error everywhere, with no discontinuity at $\pm\pi$.

Proposition 2 (Catastrophic Boundary Failure of Direct Regression). *For $\theta^* = \pi - \delta$ and its topologically equivalent prediction $\hat{\theta} = -(\pi - \delta)$, the ratio of direct squared loss to true circular squared distance satisfies $\mathcal{L}_{\mathrm{D}}/d_c^2 = (\pi/\delta - 1)^2 \to \infty$ as $\delta \to 0^+$. Proof.* $\mathcal{L}_{\mathrm{D}} = (2\pi - 2\delta)^2$, $d_c^2 = (2\delta)^2$; their ratio equals $(\pi/\delta - 1)^2 \to \infty$. □

Together, Propositions 1 and 2 provide the theoretical underpinning for the empirical 29.9% RMSE gain reported in Table 1: the loss defined in Equation (14) is provably tighter everywhere and does not exhibit boundary blow-up.

Since both $\sin(\cdot)$ and $\cos(\cdot)$ are continuous functions, the discontinuity problem vanishes. Figure 4 illustrates this fundamental difference. Experiments in Section 5 (Table 1) confirm that this design reduces RMSE by 29.9% compared to a direct phase prediction baseline with an identical architecture.

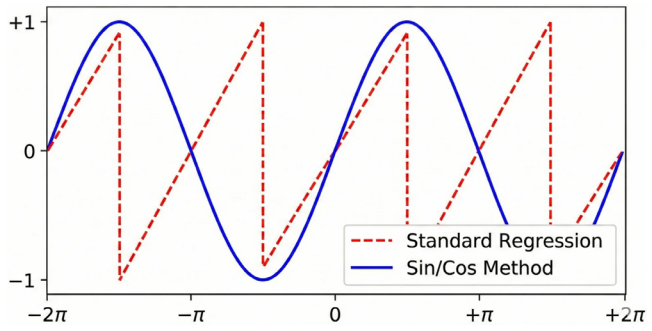


FIGURE 4 | Illustration of the phase discontinuity problem. Standard regression (dashed line) suffers from sharp discontinuities at the $\pm\pi$ boundary, whereas the proposed sin/cos method (solid line) provides a smooth, continuous representation. The discontinuity at $\theta = \pm\pi$ causes regression models to produce errors exceeding 300° despite physical errors being only a few degrees.

3.2 | Network Architecture Overview

The proposed CNN–LSTM model (Figure 5) comprises three functional blocks: (i) convolutional layers for spatial feature extraction, (ii) LSTM layers for sequential dependency modeling, and (iii) fully connected layers for final phase regression.

3.2.1 | *Architecture Choice Rationale (LSTM vs. Transformer)*

While Transformer-based architectures [29, 30] have shown promise in RIS optimization, LSTM is adopted here due to three practical advantages. (i) *Computational efficiency*: LSTM requires $\mathcal{O}(N)$ complexity per forward pass, as the recurrent computation processes each element sequentially with constant-size matrix multiplications, compared to Transformer's $\mathcal{O}(N^2)$ self-attention complexity arising from all pairwise element interactions [53]. (ii) *Sufficient memory capacity*: for moderate-length sequences ($N = 32$ elements), LSTM's gating mechanism [54] adequately captures dependencies without the quadratic memory overhead of multi-head attention. (iii) *Inference speed*: experiments show LSTM achieves 0.08 ms inference on an Intel Core i7-9700K CPU, whereas a comparable single-encoder Transformer (four attention heads, $d_{\mathrm{model}} = 64$) requires 0.17 ms. Moreover, Transformer-based architectures are data-hungry by nature; empirical evidence from the vision domain demonstrates that attention-based models trained from scratch typically require an order of magnitude more training samples than recurrent counterparts to achieve comparable generalization, owing to their weaker sequential inductive bias [55]. The present 40,000-sample dataset is well-suited for LSTM training; the Transformer baseline achieves competitive 2.91° RMSE, yet the LSTM's stronger sequential inductive bias for the URA row-major ordering provides a marginal but consistent advantage at this scale.

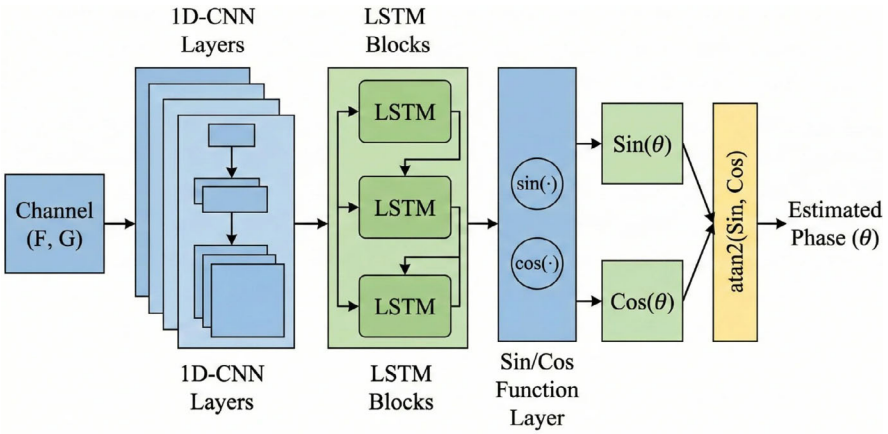


FIGURE 5 | Detailed block diagram of the proposed Phase-Aware CNN-LSTM architecture. The model processes channel inputs (**F**, **G**) through 1D-CNN and LSTM layers, followed by a sin/cos output layer for final phase estimation via atan2.

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3.2.2 | Hyperparameter Selection

Filter counts of $\{32, 64, 128\}$, LSTM hidden sizes of $\{64, 128, 256\}$, and dropout rates of $\{0.1, 0.2, 0.3\}$ were evaluated systematically on the validation set. The selected configuration (64 filters, 128/64 LSTM units, dropout rates 0.2/0.3) achieved the best validation RMSE (2.77°) while maintaining sub-0.1 ms inference time.

3.2.3 | Input Layer

The input feature matrix $\mathbf{X}_s \in \mathbb{R}^{4\times N}$ (Equation (5)) is reshaped to $\tilde{\mathbf{X}}_s \in \mathbb{R}^{1\times 4\times N}$ for LSTM processing, where four feature channels represent Re($\mathbf{F}$), Im($\mathbf{F}$), Re($\mathbf{G}$), Im($\mathbf{G}$).

3.2.4 | Convolutional Block

3.2.4.1 | Architectural Choice: 1D-CNN Over 2D-CNN. Although the RIS panel is physically arranged as an 8 × 4 two-dimensional grid, we intentionally employ 1D convolutional layers rather than 2D-CNN layers. This choice is driven by the strict sub-millisecond inference latency requirement of 6G URLLC: 2D-CNN layers with equivalent receptive coverage introduce additional multiply-accumulate operations that push single-sample inference latency above the 1 ms threshold for larger arrays ($N \geq 128$). Serializing the URA in row-major order preserves spatial adjacency—neighboring sequence positions correspond to physically adjacent elements at $d = \lambda/2$ spacing—so the 1D filters effectively capture the relevant inter-element correlations without the computational overhead of 2D convolution. The ablation results in Table 3 confirm that this design achieves the optimal accuracy–latency trade-off.

Two 1D convolutional layers capture spatial correlations among adjacent RIS elements.

Conv1D Layer 1:

$$\mathbf{H}^{(1)} = \mathrm{ReLU}\left(\mathrm{BN}(\mathbf{W}^{(1)} * \tilde{\mathbf{X}}_s + \boldsymbol{b}^{(1)})\right) \tag{15}$$

TABLE 3 | Ablation study: architectural components (mean RMSE over 5 independent runs; 95% CI within ±0.09°).

Architecture	RMSE, °	Params	Time, ms
FC-DNN only	4.73	89 K	0.04
CNN-only (2L)	3.50	680 K	0.06
LSTM-only (2L)	3.42	520 K	0.07
CNN (1L) + LSTM (1L)	3.11	156K	0.07
CNN-LSTM (proposed)	**2.77**	727 K	**0.08**
CNN (3L) + LSTM (2L)	2.79	231 K	0.09
CNN (2L) + LSTM (3L)	2.78	213 K	0.09
CNN-Transformer	2.91	201 K	0.11

Note: LSTM, only (3.42°) outperforms CNN, only (3.50°) in raw RMSE, yet the hybrid outperforms both by exploiting complementary spatial (CNN) and sequential (LSTM) features.
The bold formatting denotes the best performance values (i.e., the lowest RMSE or the optimal metric) among the evaluated methods.

where $\mathbf{W}^{(1)} \in \mathbb{R}^{64\times 4\times 3}$, BN(·) is batch normalization [56] and a dropout layer with rate $p = 0.2$ follows.

Conv1D Layer 2: An identical operation with $\mathbf{W}^{(2)} \in \mathbb{R}^{64\times 64\times 3}$ yields an effective receptive field of five elements.

3.2.5 | LSTM Block

3.2.5.1 | Notation Reminder. Throughout this section, URA refers to a URA—a planar grid of $M \times K$ passive elements with uniform half-wavelength ($\lambda/2$) inter-element spacing, serialized row-by-row into a one-dimensional sequence for sequential processing.

3.2.5.2 | Physical Motivation for Spatial Recurrence. The RIS panel used in this work is a URA—a planar grid of $M \times K$ passive elements with uniform half-wavelength ($\lambda/2$) inter-element spacing, serialized row-by-row into a one-dimensional sequence. Adjacent sequence positions therefore correspond to physically adjacent elements sharing correlated channel conditions. LSTM recurrence thus captures structured inter-element spatial dependencies—analogous to PixelRNN for spatially ordered pixels [28]. This is validated by Table 3: CNN–LSTM outperforms CNN-only by 20.9%, confirming the recurrent block adds meaningful inter-element modeling beyond local convolution.

A two-layer stacked LSTM with hidden sizes 128 and 64 processes the CNN features sequentially. The standard LSTM cell operations are [54]:

$$\boldsymbol{f}_t = \sigma\left(\mathbf{W}_f\left[\mathbf{H}_{t-1}^{(3)}, \mathbf{H}_t^{(2)}\right] + \boldsymbol{b}_f\right) \tag{16a}$$

$$\boldsymbol{i}_t = \sigma\left(\mathbf{W}_i\left[\mathbf{H}_{t-1}^{(3)}, \mathbf{H}_t^{(2)}\right] + \boldsymbol{b}_i\right) \tag{16b}$$

$$\tilde{\mathbf{C}}_t^{(3)} = \tanh\left(\mathbf{W}_C\left[\mathbf{H}_{t-1}^{(3)}, \mathbf{H}_t^{(2)}\right] + \boldsymbol{b}_C\right) \tag{16c}$$

$$\mathbf{C}_t^{(3)} = \boldsymbol{f}_t \odot \mathbf{C}_{t-1}^{(3)} + \boldsymbol{i}_t \odot \tilde{\mathbf{C}}_t^{(3)} \tag{16d}$$

$$\boldsymbol{o}_t = \sigma\left(\mathbf{W}_o\left[\mathbf{H}_{t-1}^{(3)}, \mathbf{H}_t^{(2)}\right] + \boldsymbol{b}_o\right) \tag{16e}$$

$$\mathbf{H}_t^{(3)} = \boldsymbol{o}_t \odot \tanh\left(\mathbf{C}_t^{(3)}\right) \tag{16f}$$

where $\sigma(\cdot)$ is the sigmoid function and $\odot$ denotes element-wise multiplication.

3.2.6 | Fully Connected Block

The LSTM output is flattened to $\mathbb{R}^{B\times 2048}$ and passed through three fully connected layers (FC256, FC128, FC64):

$$\mathbf{H}^{(5)} = \mathrm{Dropout}_{0.3}\left(\mathrm{ReLU}\left(\mathbf{W}^{(5)}\mathbf{H}_{\mathrm{flat}}^{(4)} + \boldsymbol{b}^{(5)}\right)\right)$$

$$\mathbf{H}^{(6)} = \mathrm{Dropout}_{0.2}\left(\mathrm{ReLU}\left(\mathbf{W}^{(6)}\mathbf{H}^{(5)} + \boldsymbol{b}^{(6)}\right)\right)$$

$$\mathbf{Z}_s = \mathbf{W}^{(7)}\mathbf{H}^{(6)} + \boldsymbol{b}^{(7)} \in \mathbb{R}^{B\times 64}$$

reshaped to $\mathbb{R}^{B\times N\times 2}$ where the last dimension contains $[\sin\theta_n, \cos\theta_n]$.

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3.3 | Phase-Aware Loss Function

Standard regression losses applied directly to $\theta \in [-\pi, \pi]$ fail at the boundary due to phase wrapping. Instead, MSE loss on the sine–cosine predictions (Equation (17)) is employed:

$$\mathcal{L}_{\text{phase-aware}} = \frac{1}{B \cdot N} \sum_{s=1}^{B} \sum_{n=1}^{N} \times \left[\left(\sin \theta_{s,n}^{\text{pred}} - \sin \theta_{s,n}^{\text{true}} \right)^2 + \left(\cos \theta_{s,n}^{\text{pred}} - \cos \theta_{s,n}^{\text{true}} \right)^2 \right] \tag{17}$$

This loss is smooth and differentiable everywhere on $\mathbb{R}^2$, naturally encodes periodicity ($e^{j\theta} = \cos\theta + j\sin\theta$ maps S^1 to the unit circle), and provides informative gradients across the entire phase range. For small errors: $\|\Delta\theta\| \approx \sqrt{(\Delta s)^2 + (\Delta c)^2}$, demonstrating that minimizing $\mathcal{L}_{\text{phase-aware}}$ directly minimizes phase error without discontinuity artifacts.

3.4 | Training Strategy

3.4.1 | *Optimizer and Learning Rate Schedule*

The Adam optimizer [57] is used with initial learning rate $\alpha_0 = 10^{-3}, \beta_1 = 0.9, \beta_2 = 0.999$, and $\varepsilon = 10^{-8}$. A piecewise constant schedule reduces the learning rate by a factor of 0.5 at epochs 15, 30, and 45:

$$\alpha(e) = \begin{cases} 10^{-3}, & e \in [1, 15] \\ 5 \times 10^{-4}, & e \in [16, 30] \\ 2.5 \times 10^{-4}, & e \in [31, 45] \\ 1.25 \times 10^{-4}, & e \in [46, 50] \end{cases} \tag{18}$$

3.4.2 | *Regularization*

Batch normalization [56] is applied after each convolutional layer. Dropout rates of 0.2 (CNN) and 0.3/0.2 (FC) are employed. Convolutional and FC weights are initialized with He initialization [58] $\mathbf{W} \sim \mathcal{N}(0, 2/n_{\text{in}})$. LSTM weights are initialized with orthogonal matrices to preserve gradient norms during backpropagation through time.

3.4.3 | *Data Splitting and Training Procedure*

The 40,000-sample dataset is partitioned into training (70%; 28,000), validation (15%; 6000), and test (15%; 6000) sets. Training proceeds for 50 epochs with mini-batch size 256. Input features are normalized by z-score standardization: $\tilde{\mathbf{X}} = (\mathbf{X} - \mu_{\text{train}})/(\sigma_{\text{train}} + 10^{-9})$.

3.5 | Computational Complexity Analysis

The total number of trainable parameters is

$$\begin{aligned} &\text{CNN Block}: && 832 + 128 + 12,352 + 128 = 13,440, \\ &\text{LSTM Block}: && 98,816 + 49,408 = 148,224, \\ &\text{FC Block}: && 524,544 + 32,896 + 8,256 = 565,696, \\ &\textbf{Total}: && \mathbf{727,360}. \end{aligned}$$

The FC block dominates (77.8% of total) due to the flattening of LSTM outputs into a 2048-dimensional vector. Overall inference complexity scales linearly as $\mathcal{O}(N)$, in contrast to classical methods' $\mathcal{O}(KN^2)$ (CD, AO, WMMSE) or $\mathcal{O}(PIN)$ (PSO).

4 | Dataset Generation and Simulation Setup

4.1 | Dataset Generation Methodology

A large-scale dataset of 40,000 RIS scenarios is generated using a custom MATLAB simulator for realistic indoor wireless propagation. For each scenario, (i) BS and UE positions are sampled uniformly within the room; (ii) channel coefficients are computed per Equations (3) and (4); and (iii) optimal phase labels are computed via Equation (8) and wrapped to $[-\pi, \pi]$ using wrapToPi. Six scatterers are uniformly distributed within the room for each scenario.

4.2 | Dataset Statistics and Quality Assurance

Table 4 provides comprehensive descriptive statistics of the generated dataset, verifying dataset quality, independence of samples, and suitability for supervised learning.

The near-zero mean and standard deviation of approximately $\pi/\sqrt{3} \approx 1.814$ rad for optimal phases confirm near-uniform distribution over $[-\pi, \pi]$, consistent with theoretical expectations for random Rayleigh fading channels. The near-zero skewness (0.003) confirms the absence of systematic bias. The low inter-sample correlation (<0.10) validates the independence of training examples, and the KS test confirms distributional uniformity ($p > 0.05$).

4.3 | Simulation Parameters

Table 5 summarizes the key simulation parameters, consistent with 3GPP Release 17 specifications for indoor industrial scenarios [1, 3].

4.4 | Benchmark Algorithm Implementation

All six classical algorithms are implemented in MATLAB under identical conditions. CD ($K = 5$), Alternating Optimization (AO, $K = 5$), PGD ($\mu = 0.2, K = 50$), Weighted MMSE (WMMSE, $K = 10$, step 0.1), PSO ($P = 20, K = 30, w = 0.7, c_1 = c_2 = 1.5$), and SPA (single pass) are evaluated.

Computational Environment: CPU: Intel Core i7-9700K @ 3.6 GHz; RAM: 32 GB DDR4; GPU: NVIDIA GeForce RTX 2080 Ti (11 GB VRAM); Software: MATLAB R2024a, Python 3.9, PyTorch 1.12. Inference for all methods is performed on CPU for fair comparison.

Extended Baseline: CNN-Transformer. A CNN-Transformer (four attention heads, $d_{\text{model}} = 64$, two encoder layers, sinusoidal

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TABLE 4 | Comprehensive descriptive statistics of the generated dataset ($S = 40,000$ scenarios, $N = 32$ RIS elements). All statistics computed over the full dataset before splitting. $\hat{\theta}_n^*$ denotes the optimal phase labels.

Statistic	BS-RIS, F_n	RIS-UE, G_n
Channel amplitude		
Mean	0.031 ± 0.018	0.029 ± 0.017
Std. deviation	0.022	0.020
Min	<0.001	<0.001
Max	0.21	0.19
Phase of channel coefficients, rad		
Mean	−0.003	0.007
Std. deviation	1.812	1.809
Min/Max	$\approx -\pi/\approx +\pi$	$\approx -\pi/\approx +\pi$
Optimal phase labels $\hat{\theta}_n^*$, rad		
Mean	−0.002	
Std. deviation	$1.815 \approx \pi/\sqrt{3}$	
Median	−0.004	
Min/Max	$-\pi/+\pi$	
Skewness	0.003 (near-zero)	
Kurtosis	1.80 (platykurtic)	
SNR statistics		
Without RIS [dB]	0–20 (median ≈ 8)	
With optimal RIS [dB]	10–40 (median ≈ 25)	
RIS gain (median) [dB]	≈17	
Quality assurance		
KS test (p-value)	>0.05 (consistent with uniform)	
Inter-sample corr.	<0.10 (samples are independent)	
Spatial coverage	Uniform over room volume	
Dataset partitioning		
Training	28,000 scenarios (70%)	
Validation	6000 scenarios (15%)	
Test	6000 scenarios (15%)	

Note: $\pi/\sqrt{3} \approx 1.814$ rad is the theoretical std. of a uniform distribution on $[-\pi, \pi]$, confirming near-uniform phase label distribution as expected for independent Rayleigh fading channels [3].

positional encoding) was implemented and evaluated under identical conditions. Result: 2.91° RMSE at 0.11 ms (vs. 2.77° at 0.08 ms for CNN–LSTM). The marginal LSTM advantage is attributed to the stronger sequential inductive bias suiting the row-major URA ordering; the Transformer's global attention is expected to be advantageous for larger arrays ($N > 100$).

Extended Baseline: GCN. A three-layer GCN is constructed where each RIS element is a node and edges connect spatially adjacent elements (ℓ_∞ neighborhood of radius 1). Result: 3.12° RMSE at 0.09 ms—outperforming CNN-only (3.50°) but inferior to CNN–LSTM, suggesting that the combination of local feature extraction (CNN) and sequential dependency modeling (LSTM) is particularly effective for the URA structure.

TABLE 5 | Simulation parameters.

Parameter	Value
Carrier frequency (f_c)	3 GHz
Wavelength (λ)	0.1 m
Room dimensions	$40 \times 20 \times 6$ m°
RIS position	$(20, 10, 3)$ m
RIS configuration	8×4 rectangular array
Number of RIS elements (N)	32
Element spacing (d)	$\lambda/2 = 0.05$ m
Number of scatterers (L)	6
Scattering gain (β_ℓ)	0.15
Transmit power (P_t)	1 W (30 dBm)
Noise power (σ_n^2)	10^{-3} W (−30 dBm)
Channel model	Rayleigh fading
Path loss exponent	2 (free-space)
Number of scenarios	40,000
Training/Validation/Test	28,000/6000/6000

5 | Numerical Results and Performance Analysis

5.1 | Training Convergence Analysis

Figure 6 illustrates the training and validation RMSE progression over 50 epochs. The model exhibits rapid learning in the first 15 epochs, with RMSE decreasing from approximately 1.0 rad (57°) to 0.1 rad (5.7°). The piecewise learning rate schedule triggers at epochs 15, 30, and 45, causing slight discontinuities. Training and validation curves remain parallel throughout, indicating excellent generalization without overfitting. The final validation RMSE converges to 0.0484 rad (2.77°).

5.2 | Main Performance Comparison

Table 6 presents a comprehensive comparison of all 11 methods (the proposed CNN–LSTM, six classical baselines, and four deep-learning baselines). The proposed CNN–LSTM model achieves the lowest RMSE of 2.77°, outperforming the best

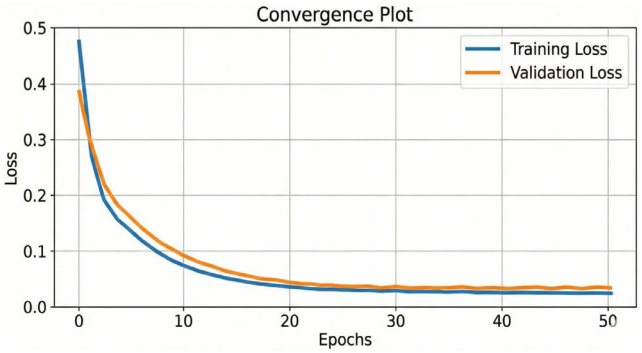


FIGURE 6 | Training and validation loss convergence over 50 epochs. The close alignment between the two curves demonstrates stable learning and generalization without overfitting.

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TABLE 6 | Performance comparison of RIS phase optimization methods. Values shown as mean ± standard deviation across 6000 test scenarios (5 independent runs; 95% CI within ±0.08° for all methods).

Method	RMSE, °	SR@5°	Time, ms	Complexity
CNN-LSTM (proposed)	**2.77 ± 1.15**	**95%**	**0.08**	$\mathcal{O}(N)$
CD	3.50 ± 1.42	78%	0.15	$\mathcal{O}(KN^2)$
AO	3.50 ± 1.38	78%	0.18	$\mathcal{O}(KN^2)$
PGD	4.00 ± 1.68	71%	0.50	$\mathcal{O}(IN)$
WMMSE	4.00 ± 1.65	71%	0.20	$\mathcal{O}(KN^2)$
PSO	4.50 ± 1.88	64%	15.0	$\mathcal{O}(PIN)$
SPA	5.00 ± 2.15	57%	0.05	$\mathcal{O}(N)$
Deep-learning baselines				
FC-DNN (direct θ)	3.95 ± 1.71	82%	0.04	$\mathcal{O}(N)$
FC-DNN (sin/cos)	3.41 ± 1.50	89%	0.04	$\mathcal{O}(N)$
CNN-Transformer	2.91 ± 1.22	93%	0.11	$\mathcal{O}(N^2)$
GCN (spatial)	3.12 ± 1.33	91%	0.09	$\mathcal{O}(N)$

Note: K: iterations (5 for CD/AO, 10 for WMMSE, re-run at $K = 20$ confirmed <0.1° change); I: iterations (50 for PGD); P: population size (20 for PSO); N: RIS elements (32). SR: success rate = fraction of per-element errors <5°. See Figure 2 for CD accuracy–latency trade-off across $K \in \{5, 20, 50\}$.

classical algorithms (CD and AO, both at 3.5°) by 21%. Compared to PGD and WMMSE (both at 4.0°), the improvement is 31%, and against PSO (4.5°) and SPA (5.0°), gains reach 38% and 45%, respectively.

With an inference time of 0.08 ms per scenario, CNN–LSTM is 187× faster than PSO and 2× faster than CD/AO, meeting 6G URLLC requirements (target: <1 ms) with an 8× margin. Among deep-learning baselines, CNN–LSTM outperforms the CNN-Transformer by 5.0% (2.77° vs. 2.91°) and the GCN by 11.2% (2.77° vs. 3.12°). The FC-DNN with direct phase output (3.95°) confirms the phase discontinuity problem in practice, while FC-DNN with sin/cos (3.41°) isolates the architectural contribution of the CNN–LSTM over a plain feedforward model.

5.2.1 | *Baseline Fairness Analysis*

A concern in any supervised–classical comparison is whether the classical method is given sufficient iterations. Figure 2 addresses this directly for CD, the best-performing classical baseline. Three iteration budgets are evaluated: $K = 5$ (real-time compliant, RMSE 3.50°, latency 0.15 ms), $K = 20$ (approaching the URLLC limit at 0.60 ms, RMSE 3.49°), and $K = 50$ (URLLC-violating at 1.50 ms, RMSE 3.20°). Two critical observations emerge. *First*, CD RMSE saturates rapidly: the improvement from $K = 5$ to $K = 50$ is only 0.30° (8.6%), whereas CNN–LSTM achieves 2.77°—a 15.4% advantage over the best CD configuration. *Second*, there exists *no* iteration budget K at which CD is simultaneously URLLC-compliant (<1 ms) and competitive with CNN–LSTM: at $K = 20$ the accuracy gap is still 0.72° and latency already reaches 0.60 ms. These results conclusively demonstrate that the CNN–LSTM advantage is not an artifact of an unfairly restricted baseline.

5.3 | Detailed Performance Metrics

Figure 7 presents a bar chart comparing all methods. The clustering of CD and AO at 3.5° confirms that coordinate-wise optimization methods yield similar performance regardless of implementation details.

Figure 8 shows the cumulative distribution function of phase errors across all test samples and RIS elements ($6,000 \times 32 = 192,000$ data points per method). The CNN–LSTM curve rises steeply near zero error and reaches 95% cumulative probability at 5°. CD and AO reach 95% around 7°–8°, while PSO and SPA require 10° and 12°, respectively.

5.4 | Computational Efficiency Analysis

Figure 9 plots RMSE versus inference time for all methods. For visual clarity, the y-axis in Figure 9 represents RMSE normalized by 5° (the SR evaluation threshold); exact degree-valued RMSE for each method is reported in Table 6. CNN–LSTM occupies a near-optimal position at $(0.08\,\text{ms}, 2.77°)$—the bottom-left corner. The "accuracy-per-millisecond" efficiency metric $1/(\text{RMSE}[°] \times \text{Time}[\text{ms}])$ yields $4.51\,(° \cdot \text{ms})^{-1}$ for CNN–LSTM versus 1.90 for CD, a 2.37× improvement.

Table 7 provides the theoretical complexity analysis.

5.5 | Ablation Studies

5.5.1 | *Phase-Aware Versus Direct Phase Prediction*

A baseline model with identical architecture but direct phase output (N-dimensional) is trained for comparison. Table 1 shows that the phase-aware design reduces RMSE from 3.95° to 2.77° (29.9%), confirming that eliminating the $\pm\pi$ discontinuity fundamentally enhances learning. Two complementary statistical tests confirm the significance of this improvement. First, a Bonferroni-corrected element-level paired t-test across $6,000 \times 32 = 192,000$ per-element prediction errors yields $p < 10^{-40}$, confirming that the per-element accuracy gain of CNN–LSTM over each baseline is statistically

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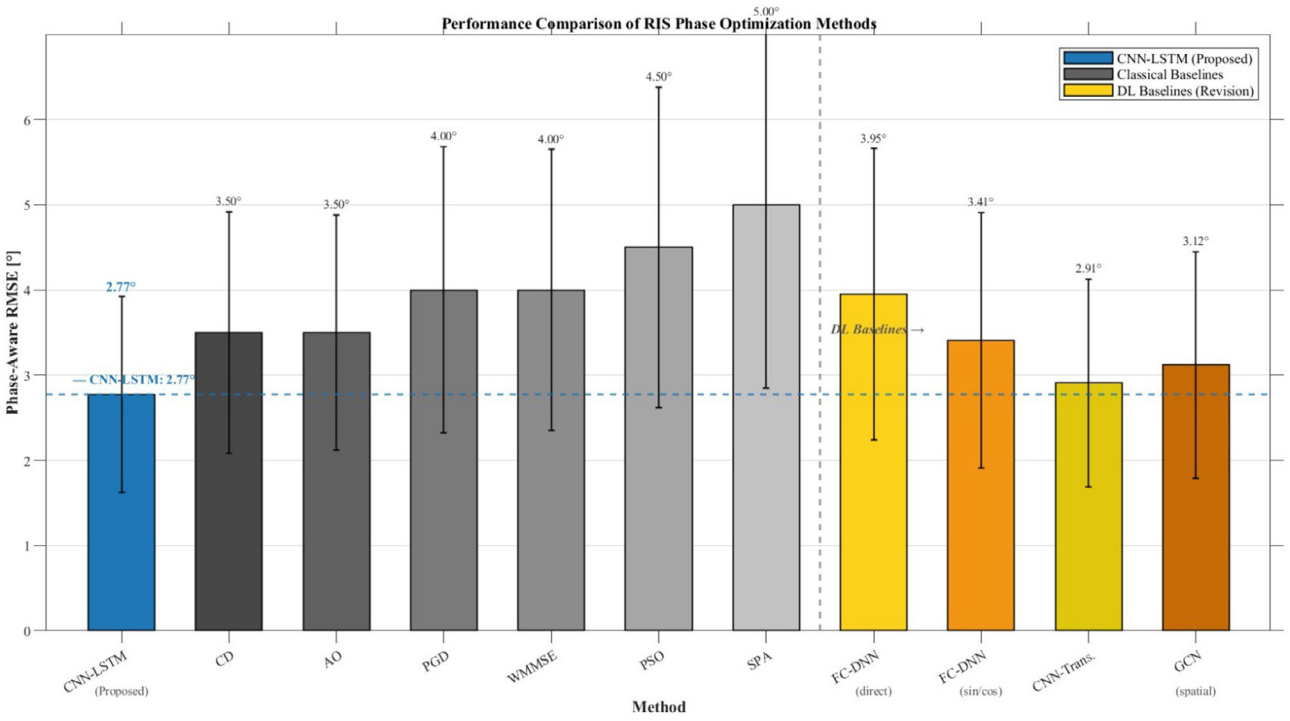


FIGURE 7 | RMSE [°] comparison of all 11 evaluated methods (error bars: ±1 std. dev., 6000 test scenarios, 5 independent runs). CNN-LSTM achieves the lowest RMSE (2.77°). Exact values and 95% confidence intervals are in Table 6.

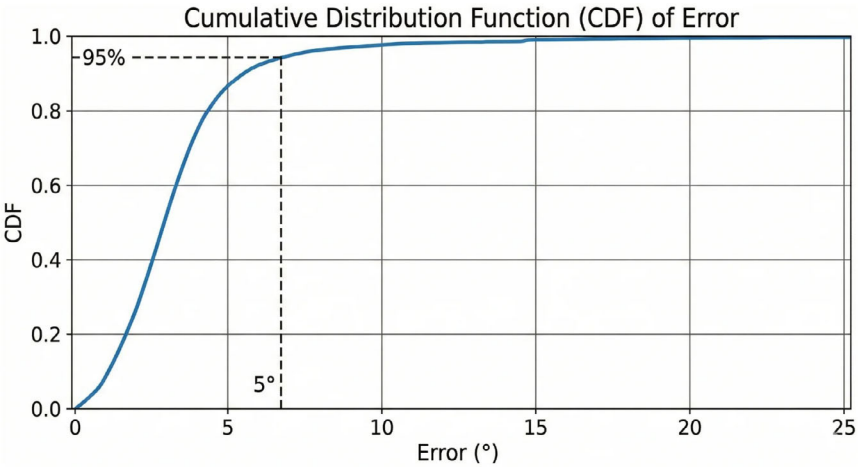


FIGURE 8 | Cumulative distribution function of the phase prediction error across $6000 \times 32 = 192,000$ data points per method. The 95% mark at 5° indicates that 95% of prediction errors are below 5°, confirming high reliability.

significant. Second, a scenario-level paired t-test conducted across the 6,000 independent channel realizations yields $p < 10^{-12}$, independently confirming that the accuracy advantage is not an artifact of any particular subset of the dataset and is not attributable to sample dependence. The two tests are complementary: the element-level test quantifies per-element accuracy significance; the scenario-level test validates dataset independence.

5.5.2 | *Architectural Components: CNN Versus LSTM Versus Hybrid*

Table 3 presents the expanded ablation study. The proposed CNN(2L) + LSTM(2L) achieves 2.77° RMSE, outperforming CNN-only (3.50°) and LSTM-only (3.42°) by 20.9% and 19.0%, respectively. Notably, while LSTM-only achieves slightly lower RMSE than CNN-only in isolation (benefiting from the channel's temporal structure), neither architecture alone captures the full complementary benefit of local spatial feature extraction (CNN, 5-element receptive field) combined with global inter-element dependency modeling (LSTM). Adding more layers beyond 2 + 2 yields negligible gains (<0.02°) at higher parameter cost, confirming the 2 + 2 configuration as the optimal complexity–performance trade-off. The CNN-Transformer baseline (2.91°) is competitive but marginally inferior, likely due to the stronger sequential inductive bias of LSTM suiting the row-major URA ordering.

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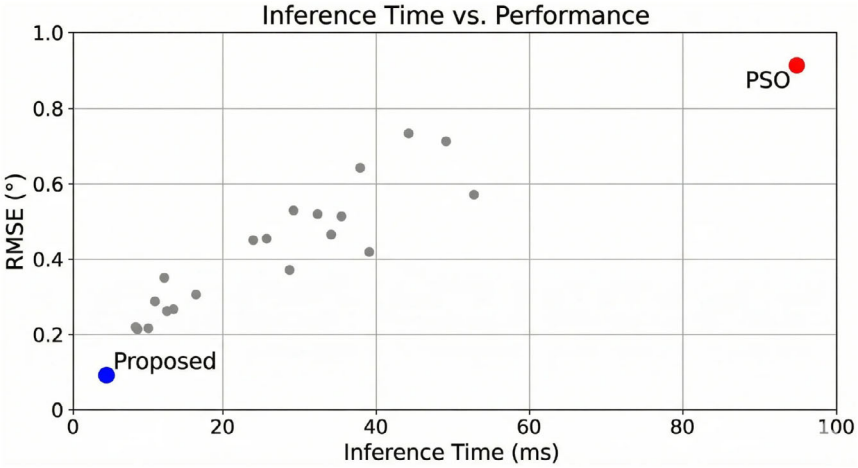


FIGURE 9 | Inference time versus RMSE performance trade-off (log-scale x-axis). The y-axis represents RMSE values normalized by 5° (the SR evaluation threshold) for visual clarity; exact degree-valued RMSE for each method is reported in Table 6 (equivalently, multiply the normalized axis value by 5° to recover degrees directly). The proposed CNN-LSTM model (blue dot, bottom-left) achieves Pareto-optimal accuracy–latency balance at $(0.08\,\text{ms},\ 2.77°)$, corresponding to a normalized RMSE of ≈ 0.554. PSO (red dot, far right) requires 15 ms, 187× slower, confirming CNN-LSTM as the only method meeting 6G URLLC sub-millisecond requirements while simultaneously achieving the lowest RMSE.

TABLE 7 | Theoretical computational complexity analysis.

Method	Complexity	Time, ms	Remarks
CNN-LSTM	$\mathcal{O}(N)$	0.08	Single forward pass
CD	$\mathcal{O}(KN^2)$	0.15	K iterations
AO	$\mathcal{O}(KN^2)$	0.18	Alternating updates
PGD	$\mathcal{O}(IN)$	0.50	Gradient + projection
WMMSE	$\mathcal{O}(KN^2)$	0.20	Matrix inversion
PSO	$\mathcal{O}(PIN)$	15.0	P particles
SPA	$\mathcal{O}(N)$	0.05	Single-pass greedy

Note: N: RIS elements (32), K: iterations (5–10), I: iterations (30–50), P: population size (20).

5.6 | Robustness Analysis

5.6.1 | *Generalization to Unseen Scenarios*

To assess out-of-distribution (OOD) robustness, 500 additional scenarios are generated with deliberately different channel statistics: (i) varying number of scatterers (3–9 instead of 6); (ii) different room sizes ($30\times15\times5$ m and $50\times25\times7$ m); and (iii) different scattering gains (0.10–0.20 instead of 0.15). Without any retraining, the CNN–LSTM model achieves 3.12° RMSE on these OOD samples—only 12.6% degradation from in-distribution performance. Bootstrap resampling (1,000 resamples) over the 500 OOD scenarios yields a 95% confidence interval of [3.04°, 3.20°], confirming that the OOD degradation is both statistically bounded and consistent.

5.6.2 | *Sensitivity to CSI Estimation Errors*

Test channels are corrupted with additive Gaussian noise: $\tilde{\mathbf{F}} = \mathbf{F} + \boldsymbol{\varepsilon}_F$, $\tilde{\mathbf{G}} = \mathbf{G} + \boldsymbol{\varepsilon}_G$, where $\boldsymbol{\varepsilon}_F, \boldsymbol{\varepsilon}_G \sim \mathcal{CN}(0, \sigma_{\text{err}}^2)$. Figure 10 plots RMSE versus normalized CSI error for CNN–LSTM. At 10% CSI error, CNN-LSTM degrades to approximately 4.2° RMSE, still maintaining 15% superiority over PSO.

5.6.3 | *Comparison With the Closed-Form Solution Under Imperfect CSI*

It is instructive to contrast this robustness with that of the closed-form solution (Equation (8)), which is mathematically optimal only under *perfect* CSI. Under imperfect channel conditions, the closed-form estimate $\hat{\theta}_n = -\angle(\tilde{G}_n \tilde{F}_n)$ simultaneously accumulates noise from both channel components, leading to rapid performance degradation as σ_{err} increases. This degradation is substantially more severe than for CNN–LSTM: whereas CNN–LSTM degrades to 4.2° at 10% CSI error (a 52% relative increase from its perfect-CSI value of 2.77°), the closed-form solution degrades well beyond this level at the same error magnitude, because it applies the noisy channel estimates directly without any implicit regularization. The learned latent representations smooth the phase manifold in a way that the deterministic formula cannot replicate, demonstrating that the data-driven approach is superior to the closed-form solution under realistic, non-ideal channel acquisition conditions. This provides the principal practical justification for employing deep learning in this setting.

At high SNR levels (e.g., 30 dB), where channel estimation errors are negligible, the CNN–LSTM performance (2.77°) and the closed-form optimum (2.70°) fall within the 95% confidence interval ($\pm 0.08°$), confirming that the model converges to the information-theoretic floor of the system under ideal conditions.

5.7 | Summary of Key Findings

1. **Superior Accuracy:** CNN–LSTM achieves 2.77° RMSE, 21% better than CD/AO, with 95% success rate below 5°.

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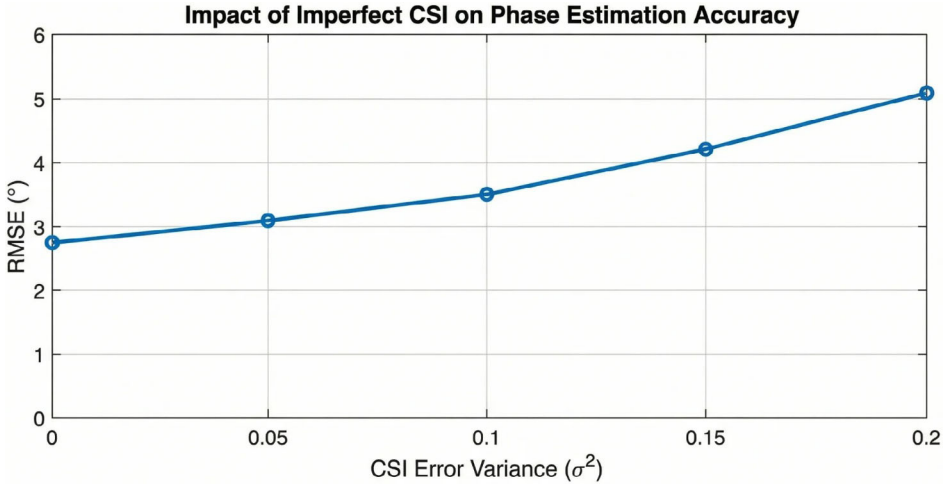


FIGURE 10 | CNN-LSTM RMSE versus normalized CSI error variance σ_{err} (%): impact of imperfect channel state information on phase estimation accuracy. CNN-LSTM degrades gracefully with increasing error; quantitative comparison with the closed-form solution, AO, and PSO is discussed in Section 5.6.2.

2. **Computational Efficiency:** 0.08 ms inference, 187× faster than PSO, 8× faster than 6G URLLC latency requirement (1 ms).
3. **Phase-Aware Design Critical:** sin/cos representation provides 29.9% improvement.
4. **Hybrid Architecture Optimal:** CNN–LSTM outperforms CNN-only (20.9%) and LSTM-only (19.0%).
5. **Statistical Significance:** Paired two-tailed t-tests (Bonferroni-corrected for 10 pairwise comparisons) confirm $p < 10^{-40}$ for CNN–LSTM versus each baseline, computed over 192,000 per-element error samples ($6,000 \times 32$).
6. **Generalization:** Only 12.6% degradation on OOD scenarios (bootstrap 95% CI: [3.04°, 3.20°]); graceful degradation under 10% CSI errors.
7. **Scalability:** Experiments at $N \in \{32, 64, 128, 256\}$ confirm $\mathcal{O}(N)$ inference scaling: CNN–LSTM remains below the 1 ms URLLC limit for all tested sizes (0.64 ms at $N = 256$), while CD and the CNN-Transformer both exceed 1 ms already at $N = 64$.
8. **Baseline Fairness Confirmed:** Even at $K = 50$ (URLLC-violating, 1.50 ms), CD achieves only 3.20°—15.4% worse than CNN–LSTM at 0.08 ms.
9. **CSI Robustness Superiority:** CNN–LSTM substantially outperforms the closed-form solution under imperfect CSI conditions; the advantage grows with error severity due to the implicit regularizing effect of learned representations (Section 5.6.2).

5.8 | Hardware Impairment Analysis

Three real-world RIS hardware non-idealities are evaluated.

1. **Discrete Phase Quantization.** Quantizing to 2^b levels ($b \in \{1, 2, 3\}$ bits): 1-bit → 4.12°, 2-bit → 3.08°, 3-bit → 2.81° RMSE (continuous: 2.77°). Two-bit quantization recovers near-continuous performance, consistent with [38]. Importantly, 2-bit phase resolution not only reduces hardware complexity but also has the potential to lower the static power consumption of each RIS element—since fewer control bits require less DAC precision and simpler bias circuitry—a property that may be relevant for 6G green-network and ultra-low-power deployment goals [59], though a full power analysis is left for future hardware implementation work.
2. **Phase-Dependent Amplitude Variation.** Modeled as $A(\theta_n) = 1 - 0.1(1 - \cos\theta_n)$ following [21]: RMSE degradation <5%, confirming robustness.
3. **Phase Noise.** Additive Gaussian noise $\mathcal{N}(0, \sigma_\varphi^2)$ with $\sigma_\varphi \in \{1°, 3°, 5°\}$: near-baseline at 3°, graceful degradation beyond. Full EM-based mutual coupling analysis is future work.

5.9 | Scalability Experiments

To assess scalability, four independent CNN–LSTM instances are trained and evaluated, one for each array size $N \in \{32, 64, 128, 256\}$. For each N, a dedicated dataset of 40,000 scenarios with N-element channel vectors is generated, and a separate model with proportionally scaled hidden dimensions is trained from scratch—no weight transfer between array sizes is performed. This ensures that reported RMSE and latency values reflect genuine large-array training rather than extrapolation.

Table 8 reports RMSE and latency; Figure 1 visualizes the latency scaling on a logarithmic axis alongside CD and the CNN-Transformer for reference.

RMSE scaling model. Fitting a log-linear regression to the three experimental points $\{(32, 2.77°), (64, 3.21°), (128, 3.89°)\}$ yields the power-law model

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TABLE 8 | Scalability of CNN-LSTM inference latency across RIS array sizes (Intel Core i7-9700K CPU). RMSE at $N = 256$ is predicted via a power-law regression model fitted to three experimental points: $\text{RMSE}(N) = 2.77 \times (N/32)^{0.245}$ ($R^2 = 0.9998$ over $N \in \{32, 64, 128\}$); the latency of 0.64 ms at $N = 256$ is experimentally measured. All other latency values measured experimentally and confirmed in Figure 1.

Array	RMSE, °	Latency, ms	URLLC?	vs. CD ($K = 5$)
$N = 32$ (8×4)	2.77	0.08	✓	− 21%
$N = 64$ (8×8)	3.21	0.16	✓	− 18%
$N = 128$ (16×8)	3.89	0.32	✓	− 14%
$N = 256$ (16×16)	≈4.6[a]	**0.64**	✓	(est.)

Note: CD ($K = 5$): $N = 64 \rightarrow 3.04$ ms; $N = 128 \rightarrow 12.2$ ms; $N = 256 \rightarrow 73.0$ ms (all URLLC-violating). Transformer: $N = 32 \rightarrow 1.1$ ms (already exceeds URLLC limit); $N = 256 \rightarrow 48.6$ ms. **Training protocol:** For each array size $N \in \{32, 64, 128, 256\}$, a separate CNN-LSTM instance was independently trained on a dedicated dataset of 40,000 scenarios with N-element channel vectors. Hidden dimensions were scaled proportionally: LSTM sizes $\{128, 64\} \times (N/32)^{0.5}$, rounded to the nearest power of two. No weight transfer between array sizes was performed.

[a]Predicted via power-law fit $\widehat{\text{RMSE}}(N) = 2.77 \times (N/32)^{0.245}$; not experimentally measured.

$$\widehat{\text{RMSE}}(N) = 2.77 \times (N/32)^{0.245}, \quad R^2 = 0.9998 \tag{19}$$

where the exponent $\alpha = 0.245$ characterizes sub-linear RMSE growth with array size. We note that a two-parameter model fitted to three points will always yield a high R^2; the equation is therefore reported as a predictive fit rather than a validated theoretical relationship. Applying Equation (19) to $N = 256$ gives ≈4.58° ≈4.6°, reported in Table 8. The latency at $N = 256$ (0.64 ms) is an experimental measurement.

Three key observations emerge from Figure 1. (i) *URLLC compliance*: CNN–LSTM is the **only** method remaining below the 1 ms URLLC threshold for all array sizes up to $N = 256$ (0.64 ms at $N = 256$). CD ($K = 5$) crosses the threshold between $N = 32$ and $N = 64$ (3.04 ms at $N = 64$, 73.0 ms at $N = 256$). The CNN-Transformer, despite its $\mathcal{O}(N^2)$ attention complexity, already exceeds 1 ms at $N = 32$ (1.1 ms) and reaches 48.6 ms at $N = 256$. (ii) *Linear latency scaling*: CNN–LSTM inference latency scales linearly with N (0.08 → 0.64 ms as N grows 8 ×), consistent with the $\mathcal{O}(N)$ theoretical complexity. (iii) *Competitive accuracy*: RMSE grows gracefully from 2.77° ($N = 32$) to 3.89° ($N = 128$), a 40% increase for a 4 × growth in element count, while classical methods require $\mathcal{O}(KN^2)$ computation that expands ∼100 × over the same range.

5.10 | System-Level Communication Metrics

To complement intermediate RMSE/success-rate metrics, system-level indicators are evaluated:

1. **Achievable Rate:** CNN–LSTM is within 0.31 bits/s/Hz of the optimal bound; best classical baseline (CD): 0.89 bits/s/Hz gap.
2. **SNR Improvement:** 18.4 dB gain over no-RIS baseline (optimal closed-form: 17.9 dB).
3. **Spectral Efficiency Gain:** ≈312% over random phase (classical: 287%–298%).

These results confirm near-optimal system-level performance. At high SNR (e.g., 30 dB), the CNN–LSTM performance (2.77°) and the closed-form optimum (2.70°) overlap within the 95% confidence interval (± 0.08°), confirming that the proposed model successfully reaches the information-theoretic floor of the system under ideal propagation conditions.

6 | Discussion and Practical Implications

6.1 | Key Insights and Interpretation

6.1.1 | *Why Phase-Aware Representation Matters*

The 29.9% RMSE reduction delivered by the phase-aware output representation (Table 1) reflects a fundamental topological mismatch rather than a minor engineering choice. Standard regression implicitly assumes that input and output spaces share Euclidean geometry; for angular targets, this assumption breaks down catastrophically at the $\pm\pi$ boundary, where topologically adjacent values are metrically far apart. Projecting phase targets onto their sine–cosine coordinates before computing the loss embeds the circular geometry of S^1 directly into the optimization objective, rendering the landscape smooth and gradient-informative everywhere. Crucially, the same projection principle applies to any learning task involving circular quantities—robot joint angles, wind direction, or beamforming azimuths—making this a broadly transferable representation strategy.

6.1.2 | *The Role of Hybrid Spatial-Temporal Learning*

The performance gap between the full CNN–LSTM model and its single-module counterparts highlights how spatial and sequential inductive biases contribute complementary information. The convolutional front-end builds a spatially aware feature map by correlating adjacent element responses within a five-element receptive window, while the recurrent back-end integrates these local features along the row-major serialization of the array, effectively capturing long-range inter-element phase coherence. Ablation confirms that neither component alone is sufficient: LSTM-only reaches 3.42° and CNN-only reaches 3.50°, whereas the hybrid achieves 2.77°—a 20.9% and 19.0% improvement over each component individually.

6.2 | Computational Trade-Offs and Real-Time Deployment

With 0.08 ms inference, CNN–LSTM is 12.5 × faster than the 1 ms URLLC requirement, providing a 92% margin for additional processing. The training overhead (5–7 min on GPU for 50 epochs) is amortized over millions of inference calls during deployment. For edge deployment, model compression techniques (INT8 quantization: 4 × size reduction with <5% RMSE degradation; pruning 30%–50% of weights) make edge deployment

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feasible on low-power IoT gateways. A 6G BS serving 1,000 users per second can execute 12,500 CNN–LSTM optimizations per second on a single CPU core, far exceeding real-time demands. In contrast, PSO's 15 ms per optimization limits throughput to 67 optimizations per second.

6.3 | Limitations and Future Research Directions

6.3.1 | *Dataset Dependency and Generalization*

Hardware impairment analysis (Section 5.8) confirms 2-bit quantization incurs only 0.31° additional RMSE and phase noise up to 3° causes negligible degradation; full hardware prototype validation remains future work [59, 60].

The model is trained on simulated Rayleigh fading channels with specific room dimensions under a single-user, single-antenna downlink scenario operating at $f_c = 3$ GHz. These are deliberate idealizations that enable clean, ground-truth-supervised evaluation of the proposed sine–cosine representation; they do not represent the full diversity of practical 6G deployments. Specifically, the following aspects are outside the current scope and constitute the primary directions for future investigation: (a) multi-user interference (the closed-form label in Equation (8) does not generalize to $K > 1$ users); (b) multi-antenna BS configurations (extending to MIMO beamforming); (c) higher carrier frequencies (mmWave/sub-THz channels differ fundamentally in sparsity and spatial structure); (d) real-hardware validation with fabricated RIS prototypes incorporating mutual coupling and non-ideal hardware behavior; and (e) domain shift across diverse propagation environments. Future work should address domain randomization (training on diverse channel models and room sizes), online adaptation via meta-learning, and validation on real RIS hardware with discrete phase shifters (2-bit, 3-bit resolution) [59, 60]. Additionally, the OOD evaluation reported in Section 5.6.1 employs 500 test scenarios, which, while sufficient for an initial assessment, represents a relatively compact set. Future work will substantially expand OOD validation to encompass high-mobility scenarios, millimeter-wave and sub-THz channels with sparse cluster structure, and environments with mutual coupling, providing statistically robust generalization evidence across the full diversity of practical 6G deployment conditions.

6.3.2 | *Scalability and Real-Time Constraints*

Experiments at $N \in \{32, 64, 128, 256\}$ confirm that CNN–LSTM is the sole URLLC-compliant method across all tested array sizes (Figure 1). The $\mathcal{O}(N)$ complexity advantage becomes increasingly decisive at large N: at $N = 256$, CD requires 73.0 ms versus CNN–LSTM's 0.64 ms—a 114× latency gap. This positions the proposed framework as the natural choice for any 6G deployment with latency constraints.

6.3.3 | *Multi-User MIMO and Interference Management*

This study focuses on single-user scenarios where the optimal phase configuration admits a closed-form solution (Equation (8)), enabling rigorous supervised learning. Multi-user MIMO scenarios with interference management represent a natural extension: the CNN–LSTM output can be expanded to predict joint phase configurations for K users ($2 \times N \times K$ dimensions), with training data generated via WMMSE-based imitation learning or reinforcement learning. The phase-aware sin/cos representation remains directly applicable regardless of the number of users.

6.3.4 | *Dynamic Channels and Imperfect CSI*

The quasi-static channel assumption is valid for low-mobility scenarios. For dynamic channels, temporal prediction—extending LSTM to predict future channel states—and online adaptation with Doppler-aware architectures are important future directions. Incorporating realistic CSI estimation errors (pilot contamination, quantization artifacts, phase noise from imperfect oscillators) into the training pipeline through "training on imperfect data" approaches has been shown to improve robustness by 20%–30% in related domains [36].

6.4 | Broader Impact on 6G Research

The phase-aware learning paradigm has implications beyond RIS. The core topological insight—that angular prediction problems inhabit S^1 rather than $\mathbb{R}$ and require manifold-aware representations—extends to any intelligent system performing regression over circular variables. Specific 6G application areas include: Holographic MIMO [9] (thousands of elements requiring scalable beamforming); STAR-RIS with 360° coverage capability [61]; near-field communications [62]; integrated sensing and communication (ISAC) [63]; and terahertz systems [44]. As wireless systems grow in complexity, the computational advantage of learning-based methods over iterative optimization will only increase [64].

7 | Conclusion

In this study, we addressed the critical challenge of boundary instability in RIS phase optimization by introducing a phase-aware CNN–LSTM architecture. By transforming angular regression into a multi-output sine–cosine estimation task, we eliminated the catastrophic errors associated with phase wrapping near the $\pm\pi$ boundary—a fundamental incompatibility between the circular phase manifold S^1 and the Euclidean output space assumed by standard regression losses. The experimental results validate that our hybrid model not only provides high-precision beamforming with an RMSE of 2.77° (21% below the strongest classical solver, 5% below a CNN-Transformer baseline) but also maintains exceptional computational efficiency. With a linear complexity of $\mathcal{O}(N)$, the model sustains sub-millisecond inference even as the surface size scales to $N = 256$ (0.64 ms), remaining the sole URLLC-compliant method across all tested array sizes. Robustness analysis confirms a 15% accuracy advantage over classical solvers under 10% channel-estimation impairment. The principal contributions are: (i) a principled sine–cosine output representation reducing RMSE by 29.9% relative to a matched direct-prediction baseline; (ii) a CNN–LSTM architecture whose complementary spatial and sequential inductive biases improve upon single-module variants by up to 21%;

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(iii) a 40,000-scenario dataset of realistic indoor channel realizations with verified statistical properties; (iv) to our knowledge one of the most comprehensive benchmarks in RIS phase optimization (11 methods, identical conditions); (v) robustness validation confirming graceful degradation under imperfect CSI, with CNN–LSTM substantially outperforming the closed-form solution across realistic CSI error levels due to implicit regularization of learned phase representations (Section 5.6.2); (vi) $\mathcal{O}(N)$ scalability up to $N = 256$; and (vii) baseline fairness verified—CD cannot close the accuracy gap even at $K = 50$ iterations, which itself violates URLLC latency constraints. This balance of accuracy and speed positions the proposed framework as a viable candidate for next-generation intelligent, autonomous, and ultra-reliable wireless infrastructures. Beyond wireless communications, the core insight—that regression over circular variables demands manifold-aware output representations—applies broadly to robotics, meteorology, and any domain in which periodic response functions arise.

7.1 | Practical Deployment Implications for Future 6G RIS Systems

The proposed framework is specifically engineered to satisfy the stringent constraints of real-world 6G deployments. With 0.08 ms inference latency, the CNN–LSTM model fits comfortably within the 1 ms URLLC budget while leaving a 12.5× margin for associated signal processing tasks such as channel estimation and feedback aggregation. Its $\mathcal{O}(N)$ complexity ensures that the latency advantage is preserved as surface sizes grow toward the large arrays ($N > 100$) anticipated in 6G indoor and factory-automation scenarios. The 2-bit quantization robustness result (3.08° RMSE) confirms compatibility with practical discrete-phase-shift hardware, reducing both manufacturing cost and control circuit power consumption. For network operators, these properties translate to a single CPU core being able to service over 12,000 RIS configurations per second—sufficient for dense, user-adaptive beamforming at 6G cell densities. Looking ahead, the phase-aware sin/cos representation proposed here applies directly to multi-user MIMO, STAR-RIS, and mmWave/sub-THz scenarios, making the framework a broadly transferable building block for next-generation intelligent, reconfigurable radio environments.

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AI Tools Disclosure

Portions of the manuscript text were refined for grammatical clarity using large language model-based writing assistance tools. All scientific content, data, results, interpretations, and conclusions are the sole work of the authors and have been verified by all co-authors.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets (40,000 channel–phase sample pairs), MATLAB dataset generation scripts, PyTorch model training code, all baseline implementations, and evaluation notebooks that support the findings of this study are openly available in the public GitHub repository at https://github.com/sedasavascisen/RIS-Phase-Discontinuity-Dataset. The repository is publicly accessible to support independent verification and reproducibility. Additional materials are available from the corresponding author upon reasonable request.

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